Data Science Toolbox Portfolio Questions

Long form questions

Daniel Lawson — University of Bristol

due Week 12

Part of Data Science Toolbox.

Deadline: Week 12 (See Timetable for details).

Data Science Portfolio Overview

- Worth 40% of the course
- Individually assessed, follow UoB coursework assessment guidelines.

What is a portfolio?

- A portfolio in Data Science Toolbox part of your ongoing assessment.
- You should attempt to work through it in time with the course material, then come back and finesse your work at the end of the Unit.
- It is composed of one section per Block.
- Each section consists of two components:
  - **Worksheets**: Multiple choice questions, submitted via **Noteable**.
    - You must answer **every** worksheet. (20%)
  - **Longform**: Deeper descriptions of material you have examined during the Worksheets and Workshop.
    - You must answer **5** of the available **8** portfolio blocks. (60%)
    - You will build evidence during class. (20%)
- The Longform content should be no more than 1 page per block (see below).

Portfolio Content

Longform content is linked **within each Block** but is also available in the Assessments page. Worksheets are found within Noteable.

Portfolio Questions:

Questions are released on a rolling basis. Check back on this document for additional content as it is released, or see Assessments for individual documents.

Generic Guidance on Portfolios

Guidance on Individual Portfolios

The Portfolio is assessed on each block from 3-10. **Blocks** 1 and 2 are marked similarly but are formative (the Formative Portfolio), i.e. do not contribute to your mark. In each block you will do two activities:

1. Multiple choice questions submitted via Noteable (log in via Blackboard). These should be straightforward, either direct from your notes or with very simple experiments you can conduct as extensions of the Workshop.
2. Long-form reflective questions that should require a deeper understanding of the course material and may require you to undertake further reading or experimentation.
    - **Preparation** that you **prepare in advance** for **in-class activity**
    - **Contributions** that you have made to your group
    - **Discussion** that you present in your portfolio

You may take the multiple-choice component at any time and it is recommended that you do this when you work through the Workshop content. The long-form content is submitted at the end of the course, and you are recommended to make a first draft/note form attempt when you first see the content, and reflect back on it in a finessing stage during the examination preparation time (in lieu of an exam).

Length and format of long-form portfolio

Your (full) Portfolio should give a **one-page** answer to your assigned question in a block. You are only required to submit work for 5 blocks. Therefore the whole Portfolio is only 5 pages long. However:

- The goal is not to make you undertake a length-finessing exercise. If the content you provide appears as if it would fit on one page after such an exercise, you can submit is anyway. **There is a strict limit of 8 pages** for the portfolio content, with answers that are clearly too long being penalised.
- You must submit **Supporting Evidence** as an appendix to the portfolio. It will consist of:
    - Your **Preparation** material, as used in class

  - Your **Contributions**, as provided to your group
  - Any additional material you have created It will not be directly assessed but may be used as evidence to support your claims, i.e. any statements you make with supporting evidence will be more favourably interpreted. This is not limited. Appropriate content is RMarkdown files knitted to pdf, Jupyter Notebooks, etc.

You will submit your portfolio as either a single PDF for all material, or a PDF plus one combined Evidence file.

For the Formative Portfolio the expectation is 1 page and the limit is 1.5 pages.

Preparing for in-class activity

You are responsible for providing advanced content to be discussed in class. You have been provided with slides which you can use after researching the context. It should cover the topic provided and cover:

- Introduction
  - Slides, workshop or other format - around 5-10 minutes to explain the idea.
  - You won't explain it in detail
  - Try to find some references that explain it properly.
- Non-trivial implication
  - Find something that is "interesting" about the topic, for example non-trivial implications, fun place it is applied, historical curiosity, etc
- Discussion activity
  - Typically questions for discussion, worksheet-like calculation to work through, or code example to explore

You are **not judged** on the professionalism of the introduction content, indeed you don't have to use them with your group - you can write on paper, use pictures or just talk. The content instead evidences your preparation and reading. Don't spend too long on this unless you intend to use it directly in your portfolio, the point is to get an interesting discussion going.

Evidencing contributions

You are expected to listen to, and respond to others. You will need to evidence these contributions when you evidence your preparation. You should be able to give 3 examples per week of suggestions you make and how it was received. You don't need to make it positive, just productive.

Portfolio content

Your preparation and in class discussion provides an opportunity to reflect about the topic as an expert, and should lead to a clear framing of a response. Your portfolio will be a 1 page explanation, consisting of approximately 1/2 page background (reference your preparation and the discussion received) and 1/2 page discussion. It should be as mathematical as is appropriate for the subject. You can relegate simple or rote content to your evidence.

Long-form Portfolio Markscheme

- The longform portfolio is worth 80% is total, so these percentages add up to 80.
  - 20%: Evidence. Simple markscheme out of 100:
    - 50 for "preparing and delivering 5 portfolios discussions" (10/discussion)
    - 30 for "contributing to at least 10 discussions" (3/discussion)
    - 20 for "ensuring that your group always received a discussion" (requires evidence that your group received discussions in every session)
    - These are evidenced in your portfolio by quoting contributions to the blackboard discussion.
    - Contributions are pass/fail. You will receive full marks for any genuine attempt.
    - For preparing a discussion, you need to evidence that you thought about it in advance by producing notes, your own slides, etc. You do not need to use the provided material, but you should work towards the given question.
  - 20%: Connection. The connection between evidence and discussion and the longform portfolio content. Uses maths project mark scheme.
  - 20%: Fit. The degree to which you address the question. Uses maths project mark scheme.
  - 20%: Depth. The degree to which you have explored and brought in advanced content. Uses maths project mark scheme.

Data Science Toolbox Portfolio Questions

03 Loss Minimisation for Decisions, Boosting, Forests

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Block 3

Portfolio 03

Portfolio Question 3.1.P - Minimising Losses for Decision Trees

- How do we stop splitting a decision tree?
  - Compare information criteria and cross validation.
  - Can you find examples of when the choice matters?

Portfolio Question 3.2.P - Behind Boosting Trees

- How are the state-of-the-art boosting algorithms implemented?
- What is the intuition behind why boosting works?

Portfolio Question 3.3.P - The best classifier

- With such a diversity of classification methods, how do we decide which is best?
  - When our test distribution is the same as our training distribution?
  - When our test data are different?
  - When we need interpretable results?

Data Science Toolbox Portfolio Questions

04 Unsupervised learning - Latent Structures

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Block 4

Portfolio 04

Portfolio Question 4.1.P - Constrained Optimisation

- How is Constrained Optimisation related to Eigenvalues and Eigenvectors?
  - Is this trick relevant to wider machine learning?

Portfolio Question 4.2.P - Mixture Modelling

- "Gaussian Mixture Modelling" theoretically dominates "soft K-means".
  - Explain Gaussian Mixture modelling and how it relates to K-means as described in the lecture.
  - Is this claim true? If so, why would we ever use K-means?

Portfolio Question 4.3.P - Dimensionality

- How would you go about making a real classifier from a clustering model, that can be deployed on a training dataset?
  - Could you use the classifier to learn the "correct" number of dimensions? How?
  - Is the dimension of all embedding methods the same?
  - Is dimension as chosen above the only concept of dimension?
  - Back up your exploration with a demonstration.

Data Science Toolbox Portfolio Questions

05 Unsupervised learning - Anomalies, p-values and Missingness

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Block 5

Portfolio 05

Portfolio Question 5.1.P - Anomaly Detection Methods

- What methods for anomaly detection are there? When should you use them?
- Give an explanation of some methods that are easily implemented in practical examples.

Portfolio Question 5.2.P - Conformal prediction

- We claimed that conformal prediction could guarantee **any** model would get calibrated probabilities!
  - We only claimed it for marginal coverage. What is meant by this? When is it useful or not?
  - What methods are available for conditional coverage?
  - What data do we need to make it work?

Portfolio Question 5.3.P - Missingness

- What conclusions can we draw from this analysis about the effect of the covariates on the outcome?
  - How do we know that imputation has done a good job?
  - If the imputation went badly, how wrong could our estimates be?
- Design an experiment to try to answer these questions, by performing an evaluation of the performance of the imputation procedure. To do this, you will need to leave further data out as cross valudation, or make adversarial imputation. Include your code and any non-essential figures as an appendix.

Data Science Toolbox Portfolio Questions

06 Perceptrons and Neural Networks

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Block 6

Portfolio 06

Portfolio Question 6.1.P - Gradient Descent

- We know that gradient descent let's us work out what direction to move in to decrease loss and improve performance. Explain backpropagation and its role in updating parameters through deep MLPs.
  - What approaches are used to improve training in practice?

Portfolio Question 6.2.P - Attention

- Explain how a transformer works (in particular, what is attention?) and compare/contrast with how a CNN and a Vision Transformer process an image.

Portfolio Question 6.3.P - Depth vs width?

- Investigate the limits of the Universal Approximation Theorem for MLPs, which says that an MLP with a single hidden layer of arbitrary width can approximate any continuous function.
- Take model_1, the first MLP from lab code and modify its architecture. First train it with 1 hidden layer with 64 neurons, compare this with 2 hidden layers with 32 each etc. for a variety of different configuration of hidden layers (make sure you have activation functions between, you might want to use a loop inside your class to make this easier to define).
- For a fixed number of epochs, record the training speed, model performance, minimum test loss etc. Produce some plots and explain your observations. You might want to watch this video for some inspiration: https://youtu.be/qx7hirqgfuU?si=45VlYxsbazvJYv3b
- Given a fixed budget of neurons, is it better to have an MLP that is wider or deeper?

Data Science Toolbox Portfolio Questions

07 Bayes and Topic Models

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Block 7

Portfolio 07

Portfolio Question 7.1.P - Bayes in Machine Learning

- How is the Bayesian approach used in machine learning?
  - Contrast simple approaches where it can be used to motivate methods, to an area where it is taken seriously.

Portfolio Question 7.2.P - Interpreting LDA

- If we've looked at a collection of documents and a model like LDA has identified some "topics", how would we interpret them?
  - You might start with perplexity and coherence.
  - Do small language models help with interpretability?
  - What about large language models?

Portfolio Question 7.3.P - Applying LDA

- Consider working with a news agency looking to understand the position of its news profile compared to "ABC news" in the workshop, and other news agencies.
  - How could you compare the datasets and why?
  - What would different choices tell you?
  - Be sure to address how evidence-based and interpretable the choices are.

Data Science Toolbox Portfolio Questions

08 Algorithms for Data Science

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Block 8

Portfolio 08

Portfolio Question 8.1.P - Complexity Classes

- What is a complexity class, and how does it relate to the broader topic of "algorithmic complexity"?
  - What does $P \neq NP$ mean and is it important?
  - How does complexity relate to security and data science?
  - How does complexity relate to quantum computing?

Portfolio Question 8.2.P - Hashing big data

- Imagine you are training an LLM and wanting to keep track of the documents its seen already, to avoid duplication and allow you to downweight topics by the proportion of computer generated content they contain.
  - How does the MinHash algorithm help solve this problem?
  - How would a sketch help?
  - Discuss the approach and any competition for the LLM problem.

Portfolio Question 8.3.P - Empirical complexity

- You benchmark an algorithm whose theoretical complexity is $O(N \log N)$, but the log-log plot of run time against N has a slope of 0 at small N, about 1 in the middle, and above 1 at large N.
  - What could cause each part of this curve?
  - When should you trust the theoretical complexity over the measurement, and when the reverse?

– How would you design a benchmark that you would trust to choose between two algorithms for a real deployment?

Data Science Toolbox Portfolio Questions

09 Parallelism

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Block 9

Portfolio 09

Portfolio 09.1.1 - Vectorisation in pytorch

- Why has modern machine learning converged on tensor operations (matrix products, convolutions, element-wise functions)?
  - Link vectorisation to automatic differentiation, and how GPUs can compute this "quickly".

Portfolio 09.2.1 - Computational Graphs

- How are complex computations structured in machine learning? What are the basic operations that make optimisation automatable?
  - Reason about the role of manual structuring of computation. When is it still useful for a data scientist to worry about it?

Portfolio 09.3.1 - Big Compute vs Big Data

- You are working on a problem for which there is a (potentially expensive) **simulator** and a (potentially inexpensive) **emulator**, which can estimate what the simulator will output, and can be updated from the simulator. You can run several simulator parameters (say 64) in parallel.
  - What choices impact how you might use the emulator to choose good parameter choices?
  - An example is given for a Random Forest. The Forest is our simulator that we are trying to model, and the regression is an emulator for its performance.

Data Science Toolbox Portfolio Questions

10 Ethics and Privacy

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Block 8

Portfolio 10

Portfolio Question 10.1.P - Opting Out of NHS Data

- Should individuals be able to opt out of their health data being used for research that benefits everyone?
  - Contrast the **individual** right to privacy with the **collective** benefit of complete data, and consider whether technical safeguards can change the answer.

Portfolio Question 10.2.P - Privacy in the Census

- Should national statistics agencies deliberately make published statistics less accurate in order to protect privacy?
  - Contrast the **privacy** guarantee given to every respondent with the **accuracy** needed by the communities who rely on the statistics, and consider who bears the cost.

Portfolio Question 10.3.P - Interpretable or Black Box?

- Should high-stakes decisions about people be restricted to interpretable models?
  - Contrast **interpretable models** with **post-hoc explanations** of black boxes, and consider what each means for detecting and challenging unfairness.

Marking Criteria

The mark ranges and descriptions in normal type below are the University of Bristol Generic Marking criteria that apply to any assessment at the University - these can be found at https://www.bristol.ac.uk/academic-quality. The descriptions in bold type are additional maths-specific criteria introduced primarily to clarify the descriptors in the case of marking maths examinations.

0-100 scale	**Criteria to be satisfied** University generic marking criteria in normal type, **Maths-specific marking criteria in bold**
100 94 89	• Work would be worthy of dissemination under appropriate conditions • Mastery of advanced methods and techniques at a level beyond that explicitly taught • Ability to synthesise and employ in an original way ideas from across the subject • In group work, there is evidence of an outstanding individual contribution • Excellent presentation • Outstanding command of critical analysis and judgement **and** • **Work develops concepts not directly presented in course material or uses known concepts to answer hard, unfamiliar questions that require calculations/methods not similar to any course material** • **An elegance of mathematical work beyond that expected for the level of the course** • **Of a quality that could be distributed to fellow students as an example of exceptional work**
83 78 72	• Excellent range and depth of attainment of intended learning outcomes • Mastery of a wide range of methods and techniques • Evidence of study and originality clearly beyond the bounds of what has been taught • In group work, there is evidence of an excellent individual contribution • Excellent presentation **and** • **On standard but unfamiliar problems, carrying out calculations with no errors of understanding** • **Demonstrates a high level of technical competence with very few mistakes of any kind** • **Great clarity in mathematical arguments**
68 65 62	• Attained all the intended learning outcomes • Able to use well a range of methods and techniques to come to conclusions • Evidence of study, comprehension and synthesis beyond the bounds of what has been explicitly taught • Very good presentation of material • Able to employ critical analysis and judgement • Where group work is involved there is evidence of a productive individual contribution **and** • **Able to make a good attempt at standard but unfamiliar problems, with some minor errors** • **Demonstrates technical competence, perhaps with some shortcomings** • **Clear mathematical arguments**

0-100 scale	**Criteria to be satisfied** University generic marking criteria in normal type, **Maths-specific marking criteria in bold**
58 55 52	• Some limitations in attainment of learning objectives, but has managed to grasp most of them • Able to use most of the methods and techniques taught • Evidence of study and comprehension of what has been taught • Adequate presentation of material • Some grasp of issues and concepts underlying the techniques and material taught • Where group work is involved there is evidence of a positive individual contribution **and** • **Able to start standard but unfamiliar problems but with significant errors** • **Able to complete competently "bookwork" questions that have been seen in the course material**
48 45 42	• Limited attainment of intended learning outcomes • Able to use a proportion of the basic methods and techniques taught • Evidence of study and comprehension of what has been taught, but grasp insecure • Poorly presented • Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete **and** • **Able to complete "bookwork" questions that have been seen in course material with few errors** • **Gaps or inconsistencies in the mathematical argument**
35	• Attainment of only a minority of the learning outcomes • Able to demonstrate a clear but limited use of some of the basic methods and techniques taught • Weak and incomplete grasp of what has been taught • Deficient understanding of the issues and concepts underlying the techniques and material taught **and** • **Able to reproduce work seen in course material, but with some errors**
7-29	• Attainment of nearly all the intended learning outcomes deficient • Lack of ability to use at all or the right methods and techniques taught • Inadequately and incoherently presented • Wholly deficient grasp of what has been taught • Lack of understanding of the issues and concepts underlying the techniques and material taught **and** • **Unable to reproduce satisfactorily even "bookwork" questions that have been seen in course material**
0	• No significant assessable material, absent or assessment missing a "must pass" component